

Vishal Yadav

Senior Software Engineer

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WZ-76/A, Todapur, New Delhi

Portfolio Website

Professional Summary

Senior Software Engineer with 9+ years of experience owning backend systems end-to-end, from architecture and design through deployment and production support. Expertise in Python, Django/DRF, AWS cloud infrastructure, Blender-based automation systems, and Generative AI application development including RAG pipelines and LLM integrations. Experienced in leading engineering teams, driving technical decisions, and taking full accountability for product reliability and delivery timelines.

Work Experience

Senior Software Engineer

Mar 2023 – Present

BotLab Dynamics, IIT Delhi

- Own end-to-end technical delivery for the company's core drone animation software, leading a team of three engineers from design through production deployment.
- Architected and delivered the image projection algorithm for drone systems, taking the feature from concept to production rollout.
- Drive the full lifecycle of the company's Blender plugin suite in Python (bpy), including design decisions, code ownership, and long-term maintainability.
- Designed and delivered the precision autonomous landing algorithm, owning correctness and reliability of drone positioning in production.
- Architected a collision-free drone transition and pathfinding system, taking ownership of its deployment and scalability on AWS EC2.
- Led design and development of a full animation generation system, owning sequence logic, velocity profiles, collision avoidance, color transitions, and timed formations end-to-end.
- Took initiative to integrate Generative AI capabilities into the workflow, including LLM API calls (OpenAI, Claude, Gemini) and Model Context Protocol (MCP) for connecting Blender tools to AI systems.

Project Assistant

Aug 2018 – Mar 2023

CSIR-National Physical Laboratory (NPL), Delhi

- Owned end-to-end development of a web-based educational game for CSIR-JIGYASA (Government of India) demonstrating time synchronization concepts.
- Independently designed and built a Python-based socket application for SATRE TWSTFT modem data extraction, benchmarking it against the existing ISRO-developed software.
- Took ownership of a Python GUI for real-time and stored data visualization, covering time offset, standard deviation, and lock percentage metrics.
- Built and maintained monthly automation software for time-offset calculations, ensuring accurate synchronization of ISRO timing with IST.
- Owned the automation of daily data retrieval and upload via Linux-based shell scripts for the NavIC Receiver and SATRE Modem.

Junior Software Engineer

Sheetal Enterprises, Delhi

Aug 2017 – Aug 2018

- Developed and maintained multiple Python-based software applications.
- Improved application performance by 13 percent through optimization and refactoring.

Education

B.Tech, Instrumentation and Control Engineering

2013 – 2017

Guru Gobind Singh Indraprastha University, Delhi

Class XII (CBSE)

2013

Midfields Senior Secondary School, Delhi

Class X (CBSE)

2010

Mamta Modern Senior Secondary School, Delhi

Honors and Achievements

- Poster presentation at URSI-RCRS 2022, IIT Indore titled Digital S.I. Traceability of Time through an API.
- Certification in Introduction to Measurement and Metrology (NPL UK), 2020.
- Certification in Measurement Uncertainty (NPL UK), 2020.
- Ranked third overall and second in the last four semesters in ICE department.
- Third University Rank Holder in eighth semester.
- Awarded Intelligent and Smart Student of the Year (2012 to 2013).
- Awarded L S Gupta Memorial Award at NPL India.

Skills

Programming and Backend: Python, Django, Django REST Framework, GraphQL, Socket Programming, Linux Shell Scripting

Generative AI: LLM API Integration (OpenAI, Claude, Gemini, DeepSeek, Mistral), Retrieval-Augmented Generation (RAG), LangChain, Vector Embeddings, ChromaDB, Ollama (Local LLMs), Model Context Protocol (MCP)

Cloud and Deployment: AWS EC2, AWS S3, AWS Lambda, Docker, Gunicorn, Nginx

Databases: PostgreSQL, SQLite

Web Technologies: HTML, CSS, Bootstrap, Selenium WebDriver

Authentication: JWT, OAuth 2.0

Tools and Others: Blender Python (bpy), Cron Jobs, Git, Slack, API Development

Working Knowledge: Kubernetes, Docker Compose, Jenkins, RabbitMQ, Redis, System Design Fundamentals